

EXPERIMENT - 10

a) Internetworking with routers in CISCO PACKET TRACER simulator:-

Design and configure a simple internetwork using a router.

The screenshot displays the configuration window for a PC named 'PC1' in the Cisco Packet Tracer simulator. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Config' currently selected. The 'IP Configuration' tab is active, showing settings for the 'FastEthernet0' interface. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 fields are filled with: IP Address: 192.168.20.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.20.1, and DNS Server: 0.0.0.0. The 'IPv6 Configuration' section also has 'Static' selected, with the Link Local Address set to FE80::260:47FF:FE93:7851. The '802.1X' section is collapsed, showing 'Use 802.1X Security' as unchecked and 'Authentication' set to 'MD5'. At the bottom left, there is a 'Top' button.

Configuration Section	Option	Value
IP Configuration	Static	
	IPv4 Address	192.168.20.2
	Subnet Mask	255.255.255.0
	Default Gateway	192.168.20.1
DNS Server		0.0.0.0
IPv6 Configuration	Static	
	IPv6 Address	
	Link Local Address	FE80::260:47FF:FE93:7851
	Default Gateway	
DNS Server		
802.1X	Use 802.1X Security	☐
	Authentication	MD5
	Username	
	Password	

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::203:E4FF:FEA6:1AE2

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
##port@01900:00#
Cisco IOS02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTL152400K5
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
248856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config-if)#
Router(config-if)#end
Router#show ip interface brief
%SYS-5-CONFIG_I: Configured from console by console

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.10.1    YES manual up          up
GigabitEthernet0/1  192.168.20.1    YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
Router#show ip interface brief

Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.10.1    YES manual up          up
GigabitEthernet0/1  192.168.20.1    YES manual up          up
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
Router#
```

☐ Top

Copy Paste

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical 963, y: 520 Root 14:37:00

Time: 00:34:47

Search for device

Copper Straight-Through

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC0	PC1	ICMP	Yellow	0.000	N	0	(ed...	(delete)	
Successful	PC0	PC1	ICMP	Purple	0.000	N	1	(ed...	(delete)	

Realtime Simulation

Time: 00:38:13

Search for device

Copper Straight-Through

Scenario 0

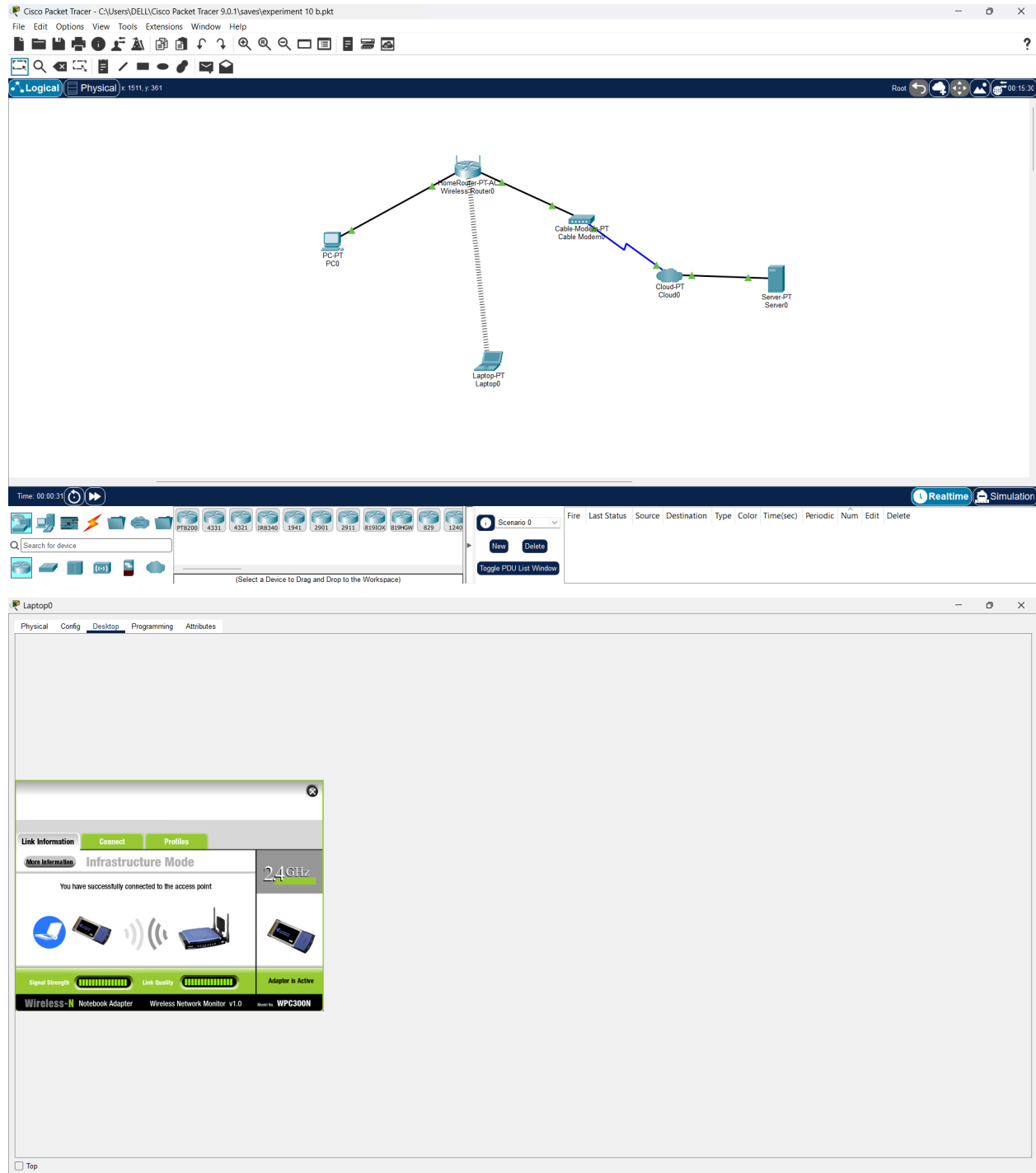
New Delete

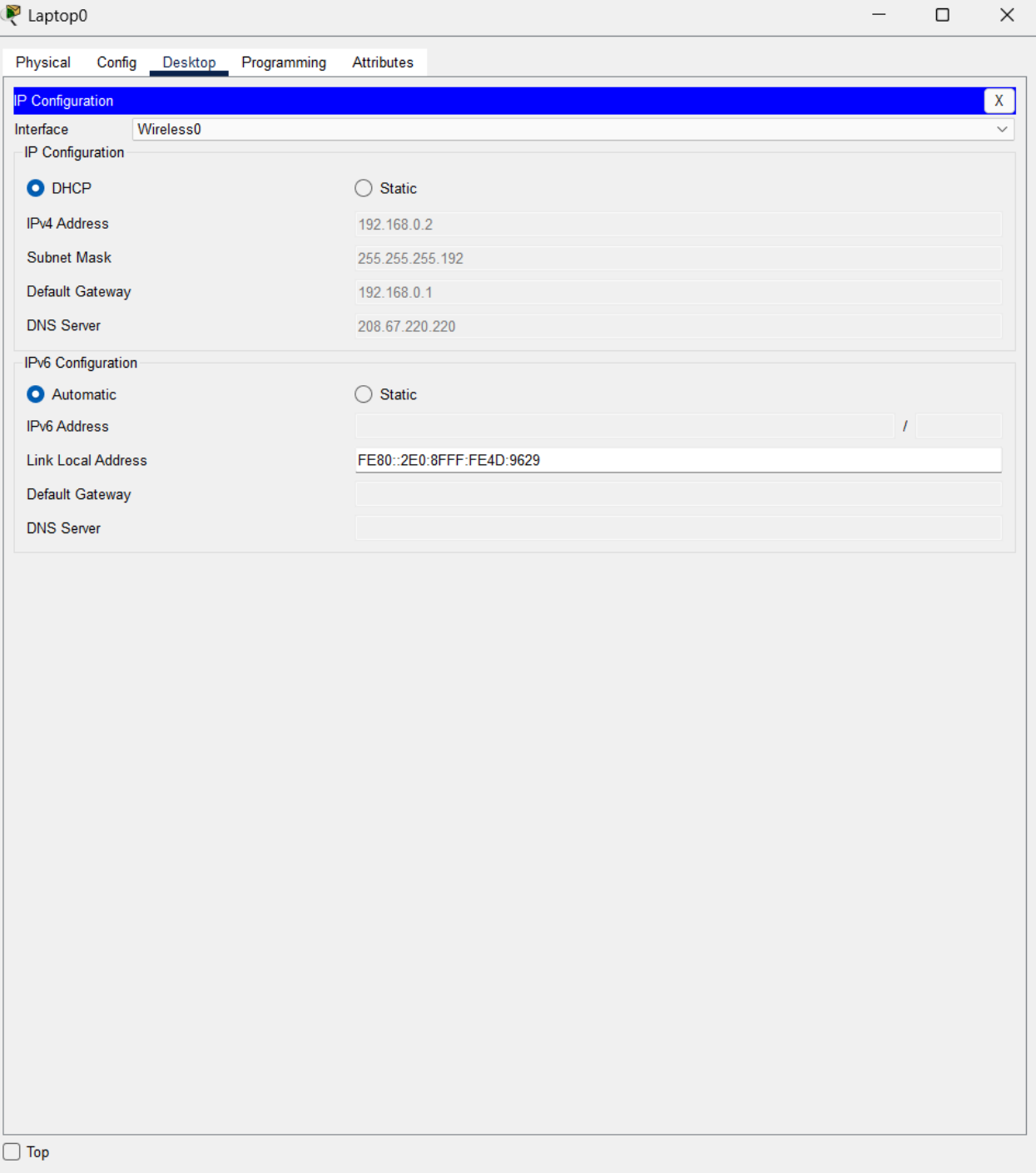
Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC0	PC1	ICMP	Yellow	0.000	N	0	(ed...	(delete)	
Successful	PC0	PC1	ICMP	Purple	0.000	N	1	(ed...	(delete)	

Realtime Simulation

b) Design and configure an internetwork using wireless router, DHCP server and internet cloud.





PC0

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

0.0.0.0

0.0.0.0

IPv6 Configuration

Static

IPv6 Address

Link Local Address

Default Gateway

DNS Server

FE80::201:64FF:FE18:C682

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

Top